

Entangled Multiplexing Transceiver

EMT

A Proposed Architecture for Faster-Than-Light Signalling via Entangled Particle Grids, Magnetic Orientation Encoding and Ensemble Weak Measurement with Full-Duplex Operation.

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25/04/2026

Original Concept Paper

<https://alternitech.square.site>

Rev.1

Abstract

This paper proposes the Entangled Multiplexing Transceiver (EMT) - a novel architecture for faster-than-light communication using entangled particle arrays.

Each EMT unit contains one slab (grid of photons held within a lattice structure). Each row of photons are magnetically aligned to the same orientation, corresponding with an alpha-numeric-symbol character on a look-up table I call the Rotary Orientation Ring.

Each slab entangled at the time of manufacture to another slab and each slab is divided into a TX half and an RX half, enabling simultaneous bidirectional (full-duplex) communication on a pair of entangled slabs. Both units are manufactured as one slab and physically cleaved apart, guaranteeing position correspondence by geometry without calibration.

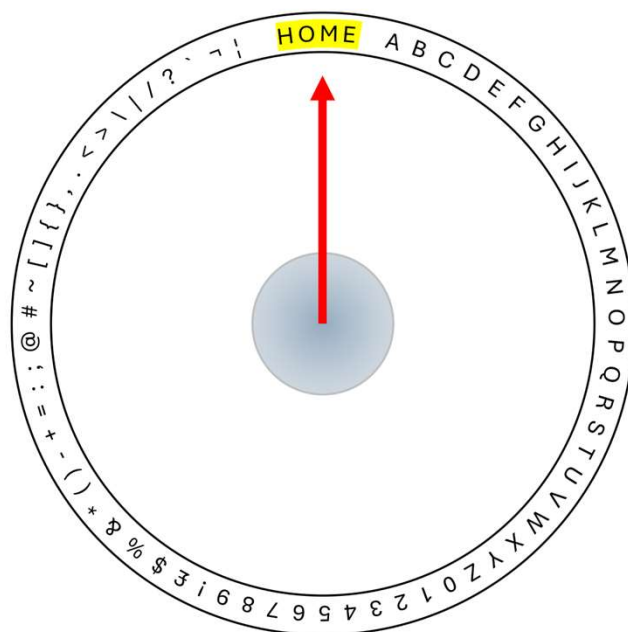
The home position at zero degrees is permanently reserved as a message delimiter. All particles in each row are set to the same orientation - the receiver averages all N readings to recover the angle with noise reduced by $\sqrt{N}$.

Orientation is written using miniature coils surrounding each photon cavity - magnetic field pulses rotate the spin state to any target angle in nanoseconds.

Orientation is read by weak laser measurement averaged by the microcontroller. The TX-to-RX row split is software-adjustable for real-time adaptive bandwidth. A single MCU handles all disc lookup, averaging, and framing.

Rotary Orientation Ring

The virtual disc is a software lookup table loaded identically onto both EMT units at manufacture, device B has its table's orientations reversed to match the spin-up/spin-down pairing of the entangled Photons. It maps each character to a unique orientation angle of the particle spin axis. The home position at exactly 0 degrees (180 for device B) is permanently reserved and is never assigned to any character. The character A begins at the first angular position immediately after the home zone.



Adding a second ring on the perpendicular vertical plane gives two independent orientation axes per particle. At ten-degree resolution this yields 1,296 unique states, sufficient for full alphanumeric and extended character coverage.

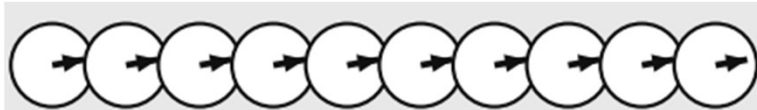


Configuration	States	Coverage
Single ring, 10°	36	A-Z + 0-9 + symbols
Dual rings, 10°	36 x 36	1,296 unique states
Dual rings, 5°	72 x 72	5,184 states
Dual rings, 1°	360 x 360	129,600 states

Multiplexing

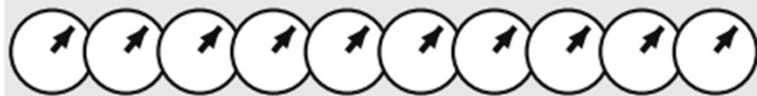
Each row of particles encodes exactly one character. Every particle in the row is set to the identical orientation angle corresponding to that character, all 10 particles per row carry orientation data. The receiver weak-measures all 10 simultaneously and averages the results, recovering the orientation with noise reduced by the square root of 10 compared to a single-particle read. This uniform orientation within each row is the core principle of ensemble noise reduction. The receiver does not need to know which particle is which — it simply averages all readings across the row and maps the result to the nearest disc position via the virtual disc lookup table (see Rotary Orientation Ring).

Row 1



H

Row 2



E

Row 3



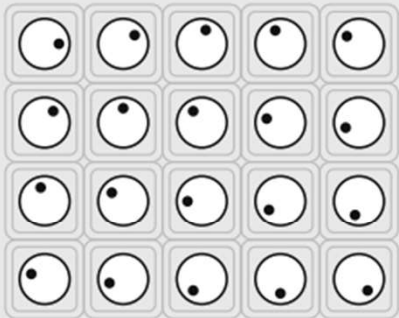
Y

Magnetic Orientation

Each photon cavity in the slab is surrounded by a miniature coil fabricated in the same electron-beam lithography step as the cavity grid. The coil generates a localised magnetic field when driven by a current pulse from the MCU. The field pulse rotates the spin state via the Zeeman effect to any target orientation angle with nanosecond precision and sub-degree accuracy. Similar to how an electric motor works. Coil diameter scales with cavity size, ranging from 50 nm for photonic crystal slabs to 200 nm for NV-centre diamond grids. Row addressing is achieved by a row-select and column-select matrix driven directly from MCU GPIO pins — no optical alignment is required for the write operation.

Miniature Coil — Per-Cavity Magnetic Write

Cavity Array — Top View



Each cavity
has its own
micro-coil

Coil diameter	~50 – 200 nm (scales with cavity)
Wire material	Superconducting NbTi or normal Cu
Field strength	1 – 100 mT (sufficient for Larmor rotation)
Pulse duration	1 – 100 ns (Rabi pi-pulse to set angle)
Addressing	Row select + column select via MCU GPIO
Fabrication	E-beam lithography — same step as cavity grid

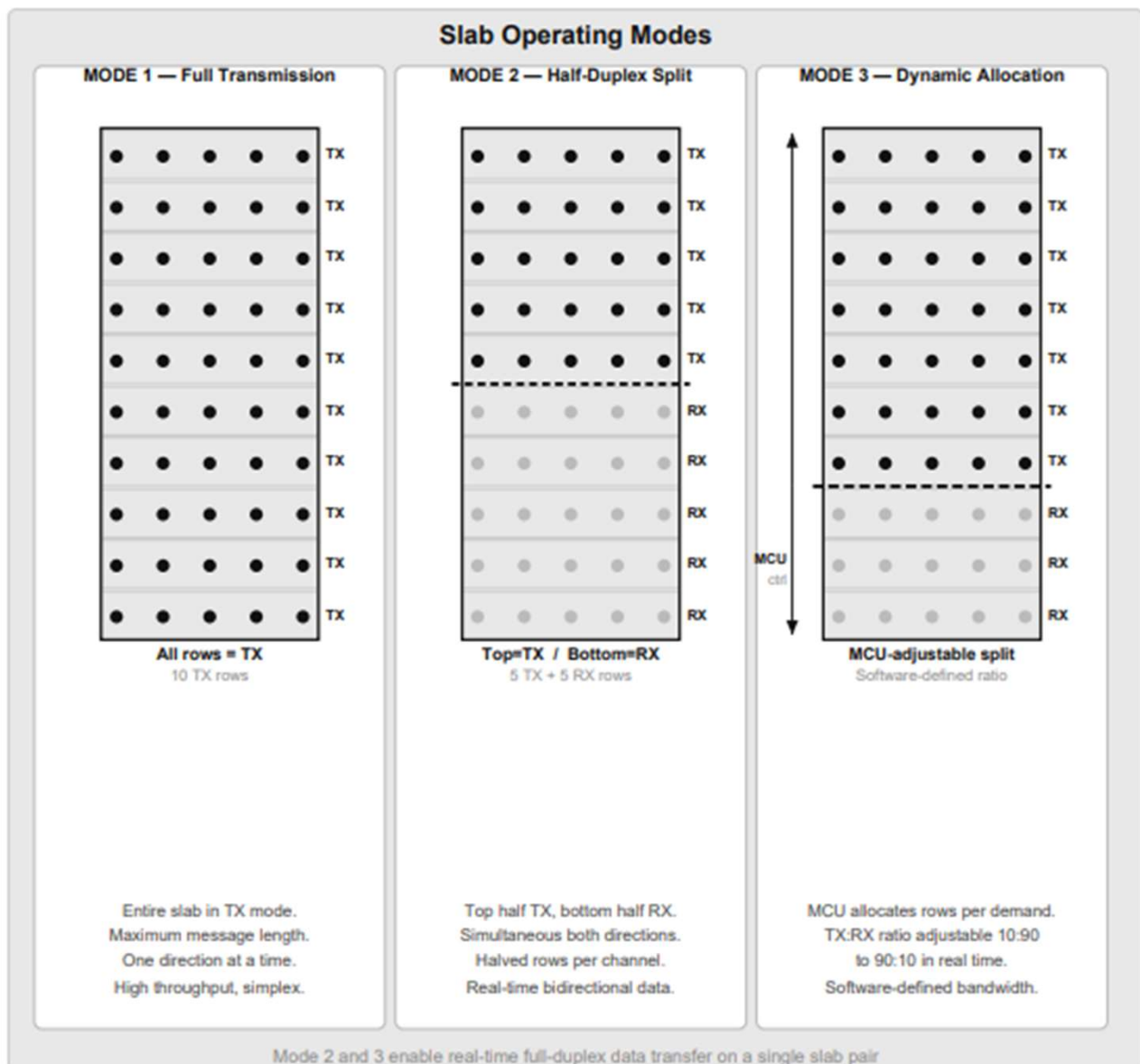
Noise Reduction

Each character row contains N particles all set to the same orientation. The MCU averages N weak measurements. Angular resolution improves as root-N. At N = 10,000 the resolution reaches 0.45 degrees — well below the 5-degree step size required for full character coverage.

Ensemble N	Angular Resolution	SNR Gain	Practical Application
1	+/-45 deg	0 dB	Single-pair proof of concept
10	+/-14 deg	~10 dB	Lab prototype
100	+/-4.5 deg	~20 dB	Research instrument
1,000	+/-1.4 deg	~30 dB	Engineering prototype
10,000	+/-0.45 deg	~40 dB	Product target
100,000	+/-0.14 deg	~50 dB	Chip-scale device

Operating Modes

The EMT slab supports three software-defined operating modes. In Mode 1, the entire slab transmits, maximising the number of characters per frame. In Mode 2, the slab is split horizontally — the top half transmits and the bottom half receives simultaneously, enabling real-time bidirectional data transfer on a single slab pair. In Mode 3, the MCU dynamically adjusts the TX-to-RX row ratio in software to match the asymmetric bandwidth demands of the application

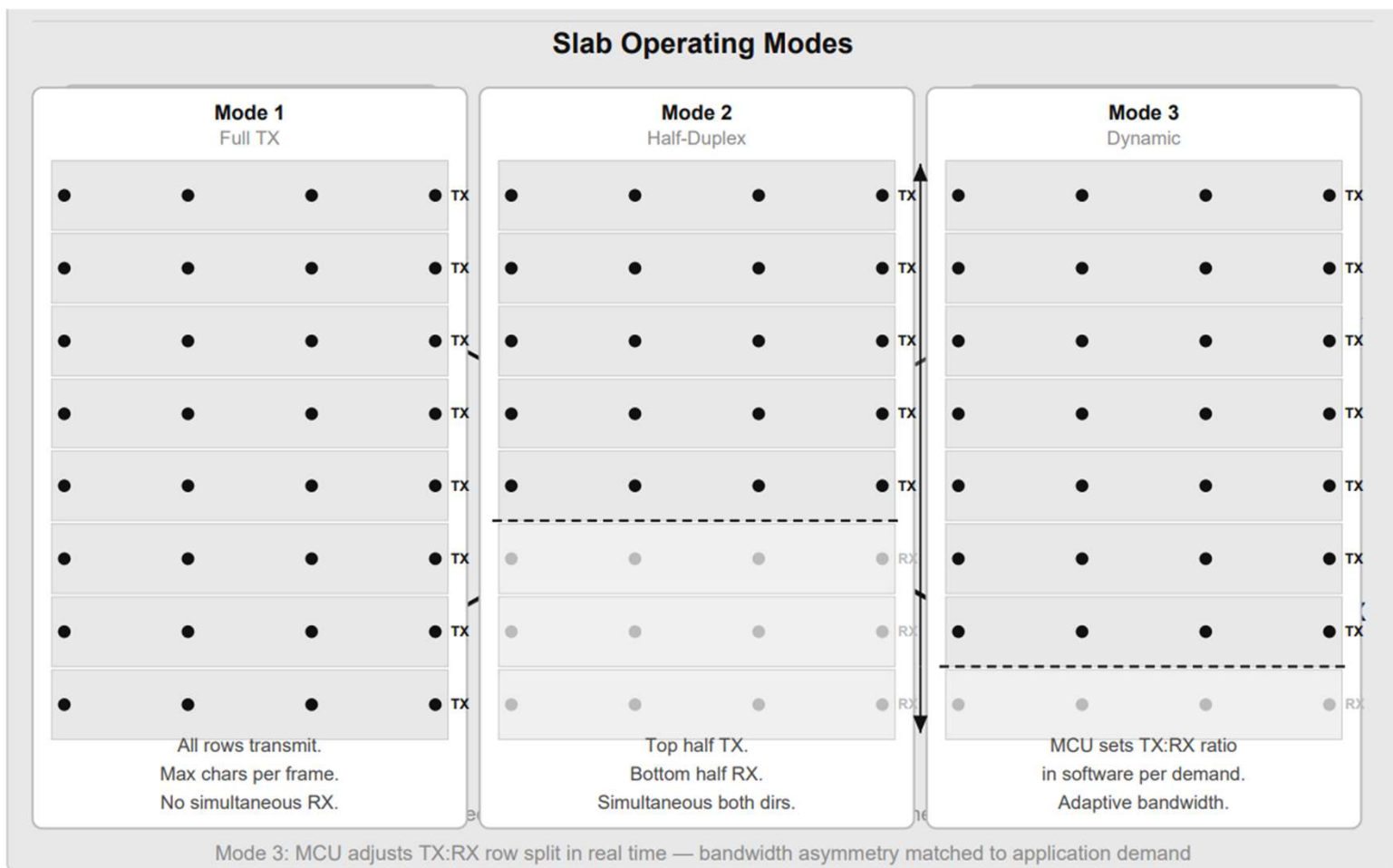


Mode bandwidth summary:

Mode	TX rows	RX rows	TX chars / frame	RX chars / frame	Use case
1 — Full TX	N	0	N	0	Burst send, no reply
2 — Half-Duplex	N/2	N/2	N/2	N/2	Real-time bidirectional
3 — Dynamic	MCU	MCU	Variable	Variable	Adaptive streaming

Full-Duplex Transceiver Architecture

Each EMT unit contains one slab. The slab is divided horizontally: the top half operates in TX mode (orienting particles to transmit characters) and the bottom half operates in RX mode (weak-reading particles to receive characters). This gives simultaneous bidirectional communication on a single slab. The entangled relationship is crossed: the TX rows of Machine A are entangled with the RX rows of Machine B, and the RX rows of Machine A are entangled with the TX rows of Machine B. Both slabs are manufactured as one unit, cleaved at the centre, and shipped to their respective sites



Reading Mechanism

Particle orientation is read using weak measurement — a low-power probe interaction that extracts partial orientation information without fully collapsing the quantum state. A probe laser illuminates the particle row; a photodetector collects the fluorescence or phase-shifted output; an ADC samples the signal; and the MCU averages N samples and maps the result to the nearest disc position.

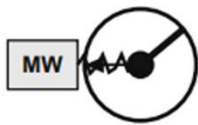
Writing Mechanism

Setting particle orientation requires a controlled rotation of the spin state. Three methods are described. The recommended production method is the miniature coil described in Section 8, which combines Method 3 (magnetic field) with per-cavity addressing and MCU-compatible drive signals

Writing Mechanism — Setting Particle Orientation

Method 1

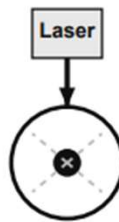
ESR / Microwave Pulse



Microwave pulse at Larmor frequency.
Pulse duration sets rotation angle.
Rabi oscillation → any angle.
Precision: $< 0.1^\circ$ achievable.

Method 2

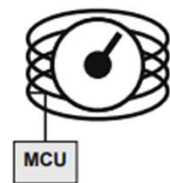
Optical Pumping



Resonant laser pulse sets spin state.
Polarisation of beam sets axis.
Nanosecond timescale operation.
Precision: $\sim 1^\circ$ per pulse.

Method 3

RF Coil (recommended)



Miniature coil per cavity.
MCU GPIO drives current pulse.
No optical alignment needed.
Row + column matrix addressing.

Method 3 recommended: miniature coil per cavity, MCU-driven, no optical alignment, nanosecond pulse precision

Manufacturing Process

Both slabs are produced as one unit and cleaved after entanglement. Each EMT unit requires two such cleavage pairs: one for its TX channel and one for its RX channel. The miniature coils are fabricated in the same lithography step as the cavity grid, adding no additional process stages.

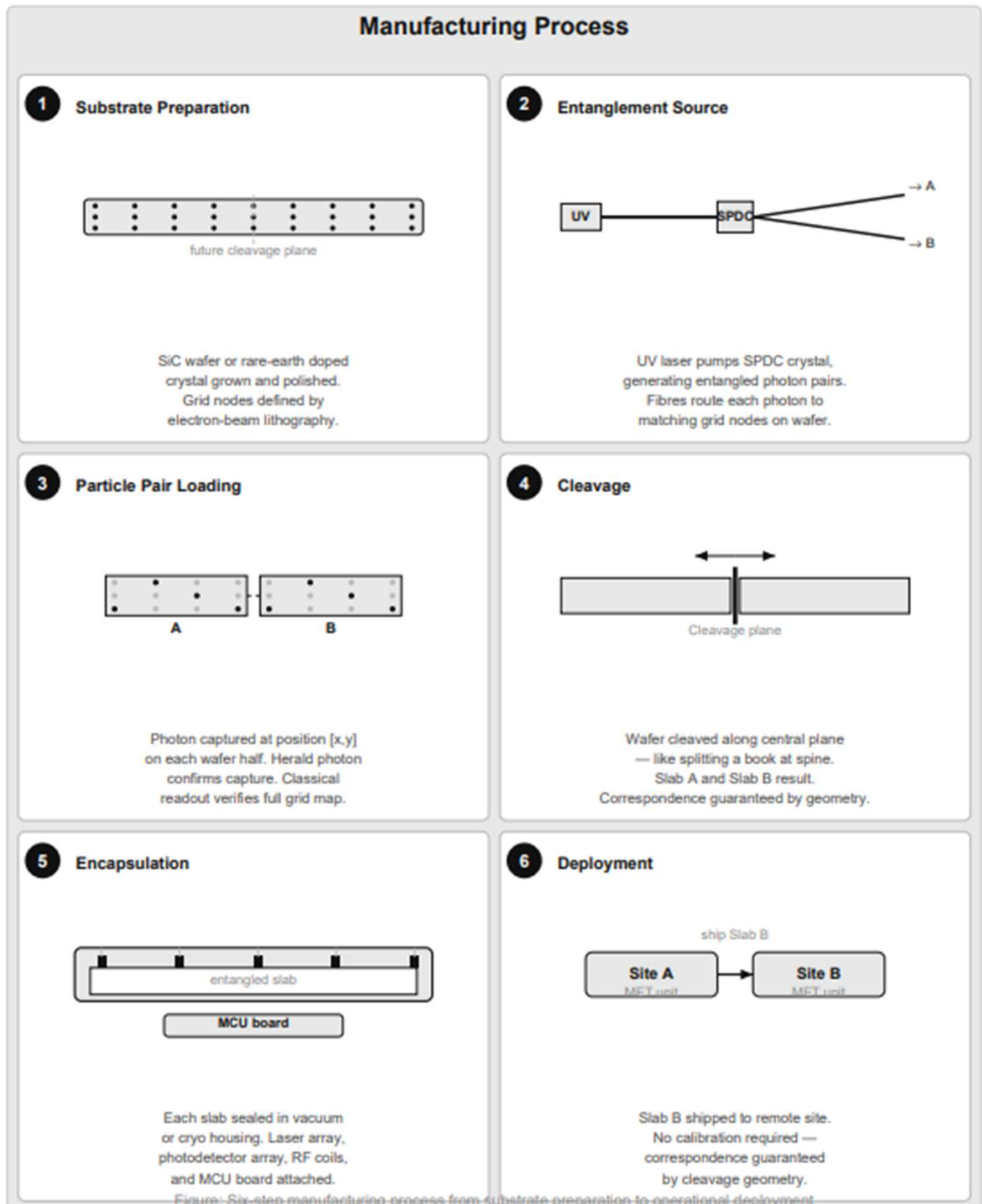


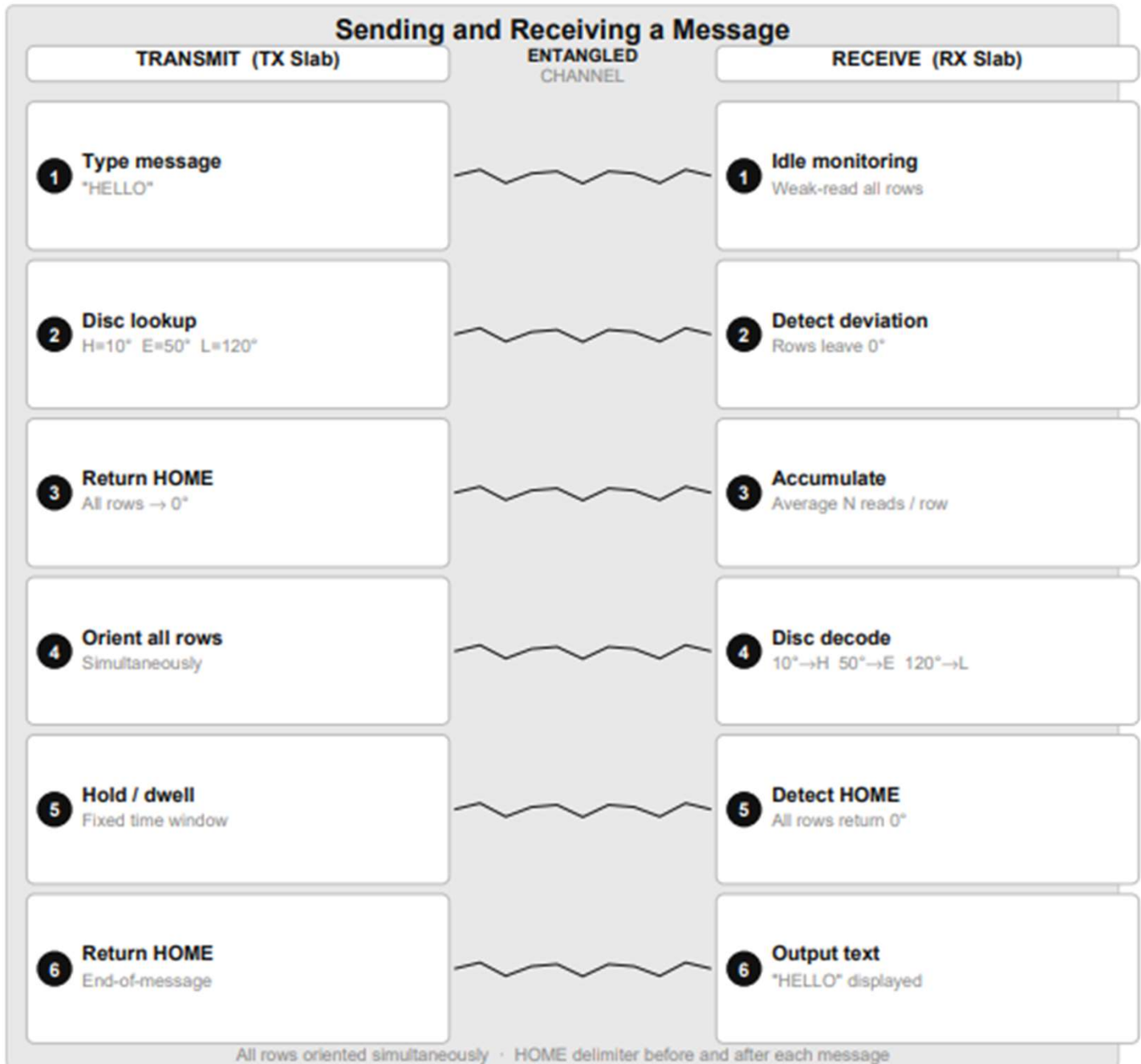
Figure: Six-step manufacturing process from substrate preparation to operational deployment

Operational Process

The EMT supports three transmission modes, each suited to different use cases. In all modes the receiver continuously weak-reads all RX rows, averages across the ensemble, and decodes via the virtual disc. HOME (0°) always serves as the start and end delimiter. Mode A — Maximum Signal (all rows, same character): All TX rows in the slab are set to the same orientation simultaneously. This maximises the total ensemble size available to the receiver for that character — if the slab has 100 rows each with 10 particles, the receiver averages 1,000 readings for a single character. This gives the highest possible signal-to-noise ratio and is used when the channel quality is poor or the angular resolution needs to be maximised. Mode B — Parallel Message (each row = one character): Each TX row is set to a different orientation corresponding to a different character in the message — all simultaneously. A 50-row slab transmits 50 characters at once. After the dwell window the transmitter loads the next 50 characters and updates all rows simultaneously again. This gives maximum throughput at the cost of per-character ensemble size. Mode C — Cascading Pipeline (rolling character advance): Rows are treated as a shift register. When row 1 has completed its dwell, it advances to the next character while row 2 is still holding its current character. This creates a continuous pipeline where the receiver is always decoding a moving window of characters. As soon as row N completes, character N+1 loads into row 1. This gives the lowest latency for streaming data at the cost of slightly reduced per-character dwell time.

Message Logistics

Title



Chip Scale Integration

Each EMT unit targets a standard chip package. The entangled slab occupies the central die area, surrounded by a miniature RF coil write array, a laser and photodetector read array, and a single microcontroller (MCU) handling all logic: ADC averaging, virtual disc lookup, message framing, and GPIO coil addressing. A Peltier cooling layer completes the stack — all within a 12 mm x 8 mm form factor. The MCU is sufficient for all real-time processing at the data rates involved.

Substrate Materials

Coherence time is the dominant factor for EMT longevity. A device at a remote site - on another continent, in orbit, or on the Moon - must maintain entanglement for the entire mission window. Microsecond or millisecond coherence limits the device to same-room distances at best.

Practical QEMT deployment requires coherence times of hours, days, or ideally weeks. The table below is ordered by FTL fitness with coherence as the primary ranking criterion.

Only the top three candidates are realistically viable for any deployment beyond the same building.

More research needs to be conducted into suitable materials for increased duration of coherence & stability.

Material	Coherence	Temp	Scalability	FTL Fitness
Rare Earth YSO (Eu/Pr)	HOURS ★★★★★	4 K	Moderate	EXCELLENT #1
NV Centres in Diamond	ms – s ★★★★★	Room temp	Limited	GOOD #2
Phosphorus in Si (Si:P)	s ★★★★★	Room temp*	Excellent	GOOD #3
Silicon Carbide (SiC)	us – ms ★★	Room temp	Excellent	MARGINAL
Trapped Ions (Ca+)	s – min ★★★	Near 0 K	Very Poor	MARGINAL (lab only)
Photonic Crystal (Si)	ps – ns ★	Room temp	Excellent	POOR — coherence fatal
Superconducting Qubits	us – ms ★★	15 mK	Good	POOR — coherence + cryo
Quantum Dots (InAs)	ns – us ★	4 K	Good	POOR — coherence short

Open Experiment Question

The no-communication theorem is proven within the current quantum mechanical formalism for discrete single-shot measurements.

The EMT architecture operates in a regime not fully addressed by existing proofs: spatially parallel transmission, continuous-variable orientation encoding, large-ensemble weak-measurement averaging, and a global phase reference shared at manufacture.

The critical experimental question is whether, at some ensemble size N , the averaged orientation signal on the receiver slab exceeds the uncertainty-principle noise floor for a controlled orientation applied to the transmitter slab.

Either experimental outcome is a significant scientific result.

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Title